# MILITARY SPECIFICATION

## SECTION, EXERCISE, TORPEDO, MARK 64 MOD 2

(For Torpedo Mark 37 Mods 2 and 3)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification describes the requirements for the manufacture, Procurements inspections and test of Exercise Section Mk 64 Mod 2, LD 547749.

## 2. APPLICABLE DOCUMENTS

* 2.1 The following-documents of the issue in effect or date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

| | |
|---|---|
| MIL-R-23811 | Recorder Assembly (For Torpedo Exercise Section Mark 64 Mod 2) |
| MIL-E-8259C | Environmental Requirements and Tests (For Torpedo Mk 37 Components and Assemblies) |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-454 | Standard General Requirements for Electronic Equipment |

DRAWINGS

Naval Sea Systems Command (Code Ident 10001)



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| | |
|---|---|
| LD 547749 | Exercise Assembly, Mk 64 Mod 2 |
| LD 547751 | Shipping (Packing) Condition Exercise Assembly, Mk 64 Mod 2 |
| LD 613311 | Preserved (Packaging) Condition Exercise Assembly, Mk 64 Mod 2 |
| 815506 | Pressure Switch |
| 1154794 | Pressure Switch |
| 1154818 | Pressure Pickup Assembly |
| 3014954 | Exercise Section Schematic Wiring Diagram |
| 3014955 | Hit Cutoff Assembled |
| 3029766 | Exercise Section Assembly |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

| | |
|---|---|
| Handbook H28 | Screw-Thread Standards for Federal Services |

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATIONS

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago, IL 60606.

AMERICAN TRUCKING ASSOCIATION, INC., PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th St. NW, Washington, DC 20036.)



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## 3. REQUIREMENTS

3.1 Description. The Torpedo Exercise Section Mark 64 Mod 2 is a torpedo shell section approximately fourteen inches long which mounts between the afterbody and the battery compartment of the torpedo. It contains all necessary programming and recording instruments for the control and documentation of a torpedo run. Included in the exercise section are a recorder, MIL-R-23811, which records torpedo data during an exercise run; an exercise timer which cuts off the drive motor and control system of the torpedo at the designated end of a run; a hit cutoff which cuts out the drive motor when the torpedo hits the target; and exercise tray which mounts all but one of the electrical parts not connected to pressure tubing; a depth control switch that terminates the run if a preset depth is exceeded; a ceiling switch that overrides the normal torpedo control circuitry and sends a "down" signal to the torpedo's elevators if the torpedo rises above a certain depth during a "not to hit" run; a noisemaker which produces an audible sound to aid in locating the torpedo after the termination of a torpedo run; and a solenoid driver assembly which pulses the noisemaker at 1 Hz. Three interconnecting cables are also considered part of the exercise section. These are the cable to the afterbody, the cable to the battery compartment, and the straight-through cable from the transducer to the control panel.

3.2 General requirements.

3.2.1 Preproduction samples. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of the exercise section is required and shall be manufactured using the methods and procedures proposed for the production. The sample shall be tested as specified in Section 4 for the purpose of determining that, prior to production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract (see 4.3.1).

3,2.2 Materials. Unless otherwise specified in the contract, requisition, or order, all materials and purchased parts entering into the final assembly of the exercise section shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 547749.

3.2.3 Finishes, coatings, construction, and dimensions. Finishes, coatings, construction, and dimensions for the exercise section shall be strictly in accordance with the applicable drawings, specifications, and publications listed in LD 547749.

3.2.4 Fungus and corrosion resistance. The exercise section and all component parts and materials shall be fungus and corrosion resistant in accordance with requirements 4 and 15 of MIL-STD-454.

3.2.5 Marking. Marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 547749.



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3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.7 Conflicting requirements. In the event of ccnflict between the documents referenced herein and this specification, the requirements of this specification shall apply.

3.3 Performance and product characteristics. The exercise section shall meet the following performance requirements and product characteristics:

3.3.1 The electrical components of the exercise section shall operate with the following power supplies:

a. AC, 115.0 ± 3.5 volts, 400 ± 8 hertz (Hz), single phase.

b. DC, 24.0 ± 2.4 volts.

3.3.2 The recorder section of the exercise section shall conform to the requirements of MIL-R-23811.

3.3.3 The exercise section shall be wired as specified in Dwg 3014954.

3.3.4 Timer. When tested as specified in 4.6.3, the exercise timer shall begin running, the oscillograph lamp shall light, and the ohmmeter shall indicate a closed circuit between terminals G and H of S31. At the end of the time interval initially set in the timer, the closed circuit between terminals G and H of S31 shall open and the noisemaker shall activate within 3 seconds, After a 20 ± 10 second interval, the timer shall cease running and the oscillograph lamp shall go off.

3.3.5 Hit cutoff. Actuation of the hit cutoff shall cause firing switch SW65 to operate. After operation of the firing switch, the closed circuit between terminals G and H of S31 shall open, when tested as specified in 4.6.4.

3.3.6 Pitch and roll, Oscillograph channels 5 and 12 shall each show a 400 Hz envelope 0.75 ± 0.50 inches (in.) wide, when tested as specified in 4.6.5.

3.3.7 Rudder. Channel 6 of the oscillograph shall deflect, and changing the position of the potentiometer arm, shall cause the channel 6 spot to move, when tested as specified in 4.6.6.

3.3.8 Port elevator. Channel 10 of the oscillograph shall deflect, and changing the position of the potentiometer arm, shall cause the channel 10 spot to move, when tested as specified in 4.6.7.

3.3.9 Starboard elevator. Channel 11 of the oscillograph shall deflect, and changing the potentiometer arm setting shall cause the channel 11 spot to move, when tested as specified in 4.6.8

3.3.10 Depth transmitter. When Depth Transmitter 1154818 is installed and 0 to 150 psig pressure is applied, channel 2 of the oscillograph shall deflect toward channel 14, and the amount of deflection shall be proportional to the applied pressure, when tested as specified in 4.6.9.



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3.3.11 Exercise ceiling pressure switch. With exercise ceiling Pressure Switch 1154794 installed, and at pressures less than 20 psig, the spot of channel 7 of the oscillograph shall be deflected 0.22 ± .12 in, toward channel "1. At pressure greater than 20 psig, the spot of channel 7 shall be in the "not energized" position, when tested as specified in 4.6.10.

3.3.12 Depth cutout pressure switch. With depth cutout Pressure Switch 815506 installed, the depth cutout pressure switch shall open at an indicated pressure of 200 ± 20 psig, when tested as specified in 4.6.11.

3.3.13 Enable and gate recording. Channel 4 of the oscillograph shall deflect 0.60 ± 0.19 in. toward channel 1, when tested as specified in 4.6.12.

3.3.14 Motor volts and rpm. Channel 7 of the oscillograph shall deflect 0.50 ± 0.25 in. toward channel 1, when tested as specified in 4.6.13.

3.3.15 Acoustic panel input. Channel 9 of the. oscillograph shall deflect 0.50 ± 0.25 in. toward channel 1, when tested as specified in 4.6.14.

3.3.16 Doppler channel. Channel 13 of the oscillograph shall deflect 1.0 ± 0.5 in. toward channel 1, when tested as specified-in" 4.6.15.

3.3.17 Course. Channel 8 of the oscillograph shall show a 400 Hz envelope 0.50 ± 0.25 in. wide, when tested as specified in 4.6.16.

3.3.18 Steering relay. Channel 3 of the oscillograph shall show a 400 Hz envelope 0.15 ± 0.06 in. wide and shall deflect 0.25 ± 0.12 toward than'nel 14, when tested as specified in 4.6.17.

3.3.19 Input level. The input level during a 30-second period shall be no more than 4 millimetres (mm) wide and spikes shall not rise over 6 mm from the bottom of the trace, when tested as specified in 4.6.18.

3.3.20 Wire command. Channel 1 of the oscillograph shall deflect 0.50 ± 0.25 in. toward channel 14, when tested as specified in 4.6.19.

3.3.21 Noisemaker. When energized with 24.0 ± 2.4 volts dc (Vdc) at a steady current flow of 0.3 to .0.6 ampere, the noisemaker shall produce maximum noise output when tested as specified in 4.6.20.

NOTE: Noisemaker may be adjusted in accordance. with Dwg 3029766.



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3.3.22 Cable continuity. The cable shall indicate continuity when tested as specified in 4.6.21.

3.4 Environmental requirements. The exercise section shall be capable of withstanding the environmental requirements and tests of this specification and shall meet all the performance requirements and product characteristics specified herein.

* 3.4.1 Vibration; operating. The exercise section shall show no damage and shall meet all the performance requirements and product characteristics of this * specification during and after the test specified in 4.4.1.

3.4.2 High temperature and humidity; nonoperating. The exercise section shall show no damage and shall meet all the performance requirements and produc characteristics of this specification after being tested as in 4.4.2.

* 3.4.3 High temperature; operating. The exercise section shall be capable of meeting all the performance requirements of this specification while undergoing the test specified in 4.4.3.

* 3.4.4 Low temperature; operating. The exercise section shall be capable o meeting all the performance requirements of this specification while undergoing the test specified in 4.4.4.

3.4.5 Operating life. The exercise head shall have an operating life of at least 30 cycles of all functions operating for 6 plus 2 minus 0 minutes, followed by a period of nonoperation of at least one hour. During each of the cycles, the unit shall be capable of meeting all the requirements of this specification (See 4.4.5).

* 3.5 Cleanup. After final assembly all exercise sections shall be thoroughly cleaned of loose, spattered or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. The equipment, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner which will insure its meeting all the requirements set forth herein. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.



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# 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the exercise section shall be classified as follows:

a. Preproduction inspection (paragraph 4.4)

b. Quality conformance inspection (paragraph 4.5)

c. Periodic production inspection (paragraph 4.4)

4.3 Sampling.

4.3.1 Preproduction sample. A preproduction sample of five exercise sections manufactured in accordance with 3.2.1 shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the exercise section by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk (See 6.2).

4.3.2 Periodic production sampling. Periodic production samples shall be randomly selected from units that have satisfactorily passed the quality conformance inspections of 4.5. These samples shall be subjected to the periodic production tests of 4.4 at an activity designated by the procuring procuring activity. Periodic production sampling shall be as follows:

Step 1. Upon initiation of production, randomly select one unit from each group of 10 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to Step 2.

Step 2. Randomly select one unit from each group of 25 units. Upon satisfactory compliance of samples selected from 3 consecutive groups, proceed to Step 3.

Step 3. Randomly select one unit from each group of 50 units. Continue Step 3 for the balance of the contract or until a defective unit is found.



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4.4 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspection of 4.5, shall be subjected to the following tests and examinations. When specified in the contract or order (See 6.2), certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

* 4.4.1 Vibration, operating. The exercise section shall be mounted securely on its joint surface without gaskets and with the entire contact surface of the joint surface supported, and subjected to the "Vibration, operating" test of MIL-E-82590. During vibration, and again following vibration, the exercise section shall be subjected to the quality conformance tests of 4.5 to verify conformance with the requirements of 3.4.1.

* 4.4.2 High temperature and humidity; nonoperating. The exercise section shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall show no damage or corrosion. After returning to room temperature, the exercise section shall be subjected to the quality conformance tests of 4.5 to verify conformance with the requirements of 3.4.2.

NOTE: Prior to testing the performance requirements, wiping or air drying may be performed. Drying by heating or baking is not allowed.

* 4.4.3 High temperature; operating condition. The exercise section shall be subjected to the "High temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall be subjected to the quality conformance tests of 4.5 to verify conformance with the requirements of 3.4.3.

* 4.4.4 Low temperature; operating condition. The exercise section shall be subjected to the "Low temperature, operating" test of MIL-E-82590 and, while at the test temperature, shall be subjected to the quality conformance tests of 4.5 to verify conformance with the requirements of 3.4.4.

* 4.4.5 Operating life. The exercise section shall be operated in accordance with this specification for a period of 6 plus 2, minus 0 minutes, then allowed to remain nonoperating for at least one hour. This cycle shall be repeated a minimum of 30 times. The exercise section shall then be subjected to the quality conformance tests of 4.5 to verify conformance with the requirements of 3.4.5.

4.5 Quality conformance inspection. Each exercise section delivered under contract shall be subjected to the tests of 4.6 and examinations of 4.7. Failure to meet any requirement specified herein shall be considered cause for rejection.



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4.5.1 Test conditions. Unless otherwise specified, the exercise section shall be subjected to quality conformance tests under the following conditions:

a. a. Temperature: Room ambient18 to 35°C (65 to 95°F).

b. Altitude: Normal ground.

c. Vibration: None.

d. Humidity; Room ambient to 95 percent relative maximum (except for dielectric test, for which specific humidity is specified).

4.5.2 Test and inspection equipment and facilities. The manufacturer shall furnish and maintain all necessary test equipment and facilities and shall provide personnel for performing all quality conformance tests. The test equipment shall be adequate in quantity and, unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required quality conformance tests.

4.5.3 Test equipment. The following items of test equipment are required to perform the quality conformance tests set forth in this specification:

a. AC power supply, 115.0 t3.**5 volts, 400 ± 8 Hz, single phase. at 2 amperes.

b. DC power supply, 24.0 ± 2.4 volts, at 5 amperes.

c. Hydraulic hand pump capable of furnishing at least 220 psig hydraulic pressure, with associated pressure tubing and fittings.

d. Pressure gage, capable of indicating from O to 220 psig, with an accuracy of 5 percent.

e. Potentiometer, 500 ohms plus or minus 10 percent, 1/2 watt or larger.

f. Resistor, 390 ohms plus or minus 10 percent, 1/2 watt or larger.

g. Resistor, 27,000 ohms plus or minus 10 percent, 1/2 watt or larger.

h. Ohmmeter, any conventional type.

i. AC transformer, 400 Hz: primary, 115 volt, secondary, 1) 6V (CT) 2) 10V

j. Resistor, 2.7 megohm plus or minus 10 percent, 1/2 watt.



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k. DC power supply, .1.25 ± 0.12 volts, 1 milliampere.

L. Ammeter capable of measuring current up to 0.6 ampere.

4.6 Quality conformance tests. Quality conformance tests shall be performed in accordance with the following procedures.

4.6.1 All pin, Plug, and connector designations referred to herein are given on Dwg 3014954.

4.6.2 Ensure that all integral components of the exercise section, as given in 3.1 and shown on Dwg 3014954, are connected as shown on Dwg 3014954.

NOTE: During tests, the positive terminal of 24 Vdc power source must be connected to the exercise shell ground.

CAUTION

Under no circumstances shall a power supply having a negative grounded output be used.

4.6.3 Timer test.

a. Connect a source of 24 Vdc between pins H and P of P23, with pin H positive. Set a time of 30 to 60 seconds on the exercise timer dial by manually turning the timer dial knob.

b. Remove the film magazine of the oscillograph and mount the oscillograph viewing screen in its place.

c. Measure between terminals G and H of S31 with an ohmmeter. It shall indicate a closed circuit.

d. Place jumpers between pins H and I of P23, pins V and L of S31, and pins M and W of S31. The exercise timer shall begin running, the oscillograph lamp shall light, and the ohmmeter shall indicate a closed circuit between pins G and H of S31. At the end of the time interval initially set on the timer in 4.6.3.a, the closed circuit between pins G and H of S31 shall open and the noisemaker shall activate Within 3 seconds. After a 20 ± 10 second interval, the timer shall cease running and the oscillograph lamp shall go off to verify the requirements of 3.3.4.

e. Disconnect the jumper between pins H and I of P23.

4.6.4 Hit cutoff test.

a. Remove the screws holding the hit cutoff assembly, Dwg 3014955, to the tray. Remove jumper between M and W of S31, and connect it between Pins j of P34 and W of S31.



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b. Reconnect the 24 Vdc source between pins H and P of P23, with pin H positive.

c. Measure between terminals G and H of S31 with an ohmmeter, which shall again indicate a closed circuit.

d. Holding the hit cutoff in one hand, strike the side of the hit cutoff with the other hand hard enough to cause firing switch SW65 in the hit cutoff to operate. After operation of the firing switch, the closed circuit between terminals G and H of S31 shall open.

e. Remove the jumpers added in 4.6.3.d. Move the negative dc power lead from pin P of P23 to terminal M of S31. This allows operation of the oscillograph without having to run the timer.

4.6,5 Pitch and roll tests.

a. Connect the outside terminals of a 500 ohm potentiometer across pins Z and a of P34, and then shunt the potentiometer with a 390 ohm resistor, connected across the same pins as the potentiometer. Leave the potentiometer center terminal (moving arm) free.

b. Apply 115 volts 400 Hz ac across pins X and Y of P34. Oscillograph channels 5 and 12 shall each show a 400 Hz envelope 0.75 ± 0,50 in, wide.

4.6.6 Rudder test. With connections as given in 4.6.5.b, connect the potentiometer moving arm to pin J of P34. Channel 6 of the oscillograph shall deflect, and changing the position of the potentiometer arm shall cause the channel 6 spot to move.

4.6.7 Port elevator test. With connections as given in 4.6.6, transfer the potentiometer moving arm from pin J of P34 to pin R of P34. Channel 10 of the oscillograph shall deflect, and changing the position of the potentiometer arm shall cause the channel 10 spot to move.

4.6.8 Starboard elevator test. With connections as given in 4.6.7 transfer the potentiometer moving arm from pin R of P34 to pin S of P34. Channel 11 of the oscillograph shall deflect, and changing the potentiometer arm setting shall cause the channel 11 spot to move.

4.6.9 Depth test. Connect the hydraulic pressure hand pump and the pressure gage to the external port of the pressure manifold, and slowly increase the applied hydraulic pressure from 0 psig to 150 psig. Channel 2 of the oscillograph shall deflect toward channel 14 and the amount of deflection shall be proportional to the applied pressure.

4.6.10 Exercise ceiling pressure switch setting.

Disconnect the ac leads from pins X and Y of P34, and connect an ohmmeter across pins j and l of P34. It shall show a closed circuit with the pressure gage reading zero pressure applied.



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b. Increase the applied hydraulic pressure with the hand pump, and adjust the setting of the ceiling pressure switch until an open circuit appears between pins j and l of P34 at an applied hydraulic pressure-of 20 ± 1 psig. The switch setting is made by turning the thumb wheel located between the switch assembly and the switch bellows.

c. Remove the ohmmeter from terminals j and l of connector P34 and apply 24 Vdc between terminals j of P34 and H of P23 (j of P34 being negative). Recheck the switch setting, made in 4.6.10.b. twice, bringing the pressure up to approximately 30 psi, each time. At pressures below 20 ± 1 psi, the spot of channel 7 of the oscillograph shall be deflected 0.22 ± 0.12 in, toward channel 1. At pressures greater than 20 ± 1 psi, the spot of channel 7 shall be in the not-energized position.

4.6.11 Depth cutout Pressure switch test.

Connect an ohmmeter across terminals G and H of S31. With hydraulic pressure applied as in 4.6.10 and with no time set on the exercise timer, there shall be an open circuit shown by the ohmmeter.

b. Manually set 30 to 60 seconds on the exercise timer. The open circuit in 4.6.11.a shall close.

c. With the hand pump, increase hydraulic pressure on the pressure port to 220 psig. The closed circuit of 4.6.11.b shall open at an indicated pressure of 200 ± 20 Psig.

4.6.12 Enable and gate recording test. Connect a 27,000 ohm resistor from the negative side of the 24 Vdc power source to pin F of P34. Charnel 4 of the oscilloscope shall deflect 0.50 ± 0.19 in. toward channel 1.

4.6.13 Motor volts and rpm test. Remove the resistor connected in 4.6.12 and connect pin K of P34 directly to the negative side of the 24 Vdc source. Channel 7 of the oscillograph shall deflect 0.50 ± 0.25 in. toward channel 1.

4.6.14 Acoustic panel input test. Connect pin P of P34 to the positive side of the 24 Vdc supply, and connect a 2.7 megohm resistor between the negative side of the 24 Vdc supply and pin N of P34. Channel 9 of the oscillograph shall deflect 0.50 ± 0.25 in. toward channel 1.

4.6.15 Doppler channel test. Connect pin W of P34 to the positive side of 24 Vdc source, and connect a 27,000 ohm resistor between the negative side of the 24 Vdc source and pin V of P34. Channel 13 of the oscillograph shall deflect 1.0 ± 0.5 in. toward channel 1.



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4.6.16 Course test. Disconnect the dc leads to P34, given in 4.6.14 and 4.6.15, and apply a 10 volt 400 Hz ac signal across pins L and M of P34. Channel 8 of the oscillograph shall show a 400 Hz envelope 0.50 ± 0.25 in. wide If deflection is incorrect, adjust potentiometer R74 until deflection is correct.

4.6.17 Steering relay test.

a. Transfer the ac input leads connected in 4.6.16 to pins C and D of P34. Channel 3 of the oscillograph shall show a 400 Hz envelope 0.19 ± 0.06 in. wide.

b. Remove the ac input lead from pin D of P34 and connect it to pin E of P34. The ac envelope given in 4.6.17.a shall disappear, and channel 3 of the oscillograph shall deflect 0.25 ± 0.12 in. toward channel 14.

c. Jumper between pins D and E of P34. The deflection given in 4.6.17.b shall remain, and in addition the ac envelope given in 4.6.17.a shall reappear.

4.6.18 Input level test. Set timer for a 30 second period. With loaded film magazine installed, apply power (24 Vdc to timer and oscillograph, and 115 volts ac 400 Hz to recording panel) and run for a 30-second period. Disconnect power, and remove and develop film. Random spikes which can be attributed to shop equipment noise should be ignored. The exercise section shall meet the requirements of 3.3.19.

4.6.19 Wire command. Connect the negative side of the 24 Vdc source to pin e of P34. Channel 1 of the oscillograph shall deflect 0.50 ± 0.25 in. toward channel 14.

4.6.20 Noisemaker test. With no time on the exercise timer, apply 24 Vdc to pins T (positive) and P (negative) of P23. The current shall be within the range of 0.3 to 0.6 ampere, and the noisemaker shall produce maximum noise output to meet the requirements of 3.3.21.

NOTE: Noise output may be adjusted in accordance with Dwg 3029766.

4.6.21 Cable continuity. Disconnect all electrical and hydraulic leads, and make cable continuity checks as given in Table I below:

Table I

| From (pin and plug) | To (terminal and plug) |
|---|---|
| A-S24 | A-S1101 |
| B-S24 | B-S1101 |
| C-S24 | C-S1101 |
| D-S24 | D-S1101 |
| E-S24 | E-S1101 |



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Table I (Cont'd)

| From (pin and plug) | To (terminal and plug) |
|---|---|
| A-P23 | A-S31 |
| B-P23 | B-S31 |
| C-P23 | C-S31 |
| D-P23 | D-S31 |
| E-P23 | E-S31 |
| F-P23 | F-S31 |
| I-P23 | I-S31 |
| M-P23 | S-S31 |
| N-P23 | K-S31 |

NOTE: All tests shall indicate continuity.

4.7 Quality conformance examinations.

4,7.1 Visual and dimensional examination. Each exercise section shall be examined to verify conformance with the requirements for workmanship and dimensions.

4.7.2 Packaging, packing, and marking. The packaging, packing, and marking shall be examined to verify conformance with the requirements of Section 5

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A, B, or C, as specified (See 6.2).

5,1.1 Levels A and B. Exercise sections shall be individually protected and unit packaged in accordance with LD 613311.

5.1.2 Level C. Exercise sections shall be afforded preservation and packaging in accordance with the manufacturer's normal shipment procedures.

5.2 Packing. Packing shall be level A, B or C, as specified (See 6.2).

5.2.1 Levels A and B. Exercise sections, packaged as specified, shall be packed in accordance with LD 547751.

5.2.2 Level C. Exercise sections, packaged as specified, shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Normal markings. In addition to any special marking required by the contract or order, unit packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.



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5.3.2 Serial number. Unit packages and shipping containers shall be marked with the serial numbers of the units contained within.

6. NOTES

* 6.1 Intended use. The exercise sections are intended for use in Torpedo Mk 37 Mods 2 and 3.

* 6.2 Ordering data. Procurement documents should specify the following:

6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Any exception to this specification.

c. Required level of preservation and packaging (See 5.1)

d. Required level of packing (See 5.2)

e. When a preproduction sample is not required (See 3.2.1 and 6.3)

f. Assigned activity for preproduction and periodic production inspections (See 4.3).

g. That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.2.2 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

| Paragraph | Data Requirement | Applicable DIDs |
|---|---|---|
| 4.4 | Certified inspection data | |

DIDs (Data Item Description/DD Forms 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (number) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

* 6.3 First article sample waiver. First article samples submitted and approved on a recent contract may be accepted by the procuring activity in lieu of an additional first article inspection. When the first article sample is waived (see 6.2.1.e) the procurement document should contain a statement specifying that the standards of workmanship exhibited by the previously approved first article sample shall determine the minimum requirements of the current contract or order.



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6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned *to* evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS
Project No. 1355-N365



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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-S-23809C(OS), Section, Exercise, Torpedo, Mk 64 Mod 2

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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